

Procedural Modeling of Crystal Clusters

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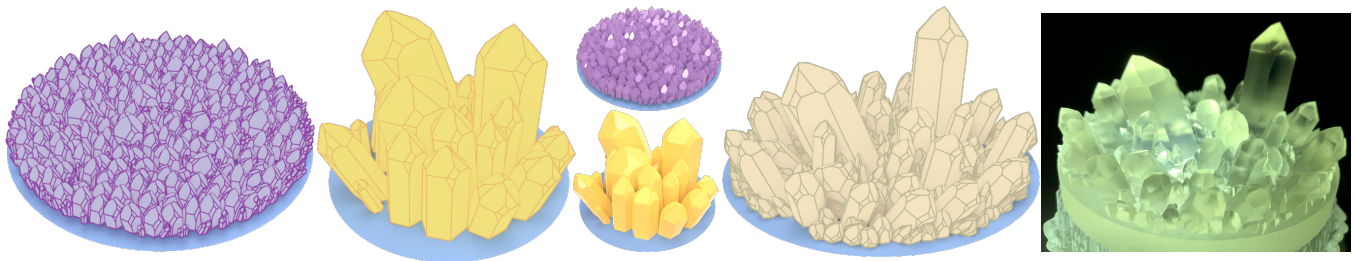


Figure 1: We generate the clusters of amethyst (purple), citrine (yellow), and crystal (beige) using our interactive procedural modeling system and fabricate the crystal cluster (right). We can confirm the clusters' visual plausibility from the rendering results with the illustrative shadings and the specular reflections.

ABSTRACT

We propose procedural modeling of crystal clusters. Based on crystallography, we model single crystals and then distribute single crystals on a base rock using a hierarchical sampling approach. Users can interactively generate various crystal clusters, including amethyst and citrine clusters. Furthermore, our approach offers a high-level specification, i.e., target shape so that the generated clusters approximate the shape; such controllability is crucial for realizing artists' intentions. Additionally, we focused on fabrication-oriented procedural modeling. We identify the possible parameter spaces based on physical experiments, i.e., test the removability of resin casts from silicone molds produced by the 3D printed crystal clusters for making replicas.

CCS CONCEPTS

• Computing methodologies → Shape modeling.

KEYWORDS

Procedural Modeling, 3D Printing

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1 INTRODUCTION

Procedural modeling is becoming popular as several 3D modeling software (e.g., Blender and Houdini [SideFX 2022]) strongly support procedural modeling workflow. The advantages of using procedural modeling are that we can generate various models by changing parameters and hold a reproducibility due to its invertibility. We propose an interactive procedural modeling system of crystal clusters in this study. We modeled single crystals based on crystallography using the Miller–Bravais index [Donnay and Le Page 1978]. Our hierarchical sampling approach enables the visually plausible formation of crystal clusters. Additionally, we support the high-level specification, i.e., target shapes (Fig. 3) inspired by [Talton et al. 2011]. Since we implement our system on a Houdini, users can interactively generate, edit, and preview the clusters in real-time. The created models are ready for 3D printing. We provide three contributions:

- We propose a crystallography-based modeling method, which differentiates our modeling from existing simple and obvious modeling methods.
- We propose a hierarchical sampling approach, which can create visually plausible crystal clusters with proper distribution density.
- We propose fabrication-oriented procedural modeling, which considers the feasibility (e.g., removability when resin casting) by identifying suitable parameter ranges, i.e., feasible parameter spaces.

2 OUR APPROACH

We propose a two-step generation method (Fig. 2). First, we generate single crystals and then distribute the single crystals for each sampling point on a base rock with parameter perturbations to obtain various cluster results.

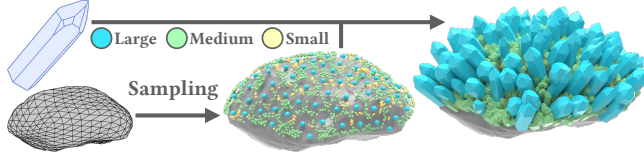


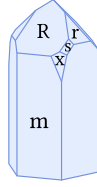
Figure 2: Method overview.

Table 1: Miller-Bravais indices of left crystals.

face	m	R	r	s	x
index(hkil)	(10 $\bar{1}$ 0)	(10 $\bar{1}$ 1)	(01 $\bar{1}$ 1)	(11 $\bar{2}$ 1)	(51 $\bar{6}$ 1)

Note: Here, $\bar{i}$ represents $-i$ and $i=-h-k$.

Modeling single crystals. We model a hexagonal crystal geometry based on the Miller-Bravais index [Donnay and Le Page 1978]. The crystal structure consists of m, R, r, s, and x-faces (inset). Table 1 lists the indices we used. We compute the face normals based on these indices. Since the actual growth of each face depends on temperature, pressure, and mineral content, we allow the user to control the face area's growth rate via simple slider components implemented in our user interface.



Modeling clusters. We propose a hierarchical sampling method to create crystal clusters on a base rock surface (Alg. 1). We empirically categorize crystals into three sizes; large, medium, and small to generate plausible models. We control the sampling density based on these categories to avoid large overlaps. From observation, we permit partial overlaps of crystals. We create a single crystal for each sampling point $p_i \in \mathcal{P}$. We perturb the growing directions within a maximum angle $\theta_{\max}$. $\theta_{\max}$ should be carefully determined to ensure the removability of resin casts from silicone molds created by the 3D printed crystals. To obtain a cluster, we generate crystal instances from large to small order and merge all crystals. We generate a cluster that approximates $\mathcal{M}$ if target shape $\mathcal{M}$ is provided. Full parameter descriptions are found in the supplemental material.

3 RESULTS

We modeled the clusters of amethyst, citrine, and crystal using our interactive procedural modeling system implemented employing the Houdini VEX script [SideFX 2022]. Our approach can generate various types of crystal clusters, as demonstrated in Fig. 1. Furthermore, we fabricated the generated model using a 3D printer with clear resin (Fig. 1 right) and confirmed that the result looks plausible. Figure 3 demonstrates our experiments on the targeted cluster generation. Fig. 3 top illustrates the successfully approximated case. However, it is difficult to approximate highly overhung portions of the target shape, as demonstrated in Fig. 3 bottom. Such target-driven procedural modeling is useful when designing objects with the artists' intentions. We can make replicas by resin casting using silicone molds created from the 3D printed crystal clusters. We created silicone molds of twin crystals with different diameters, lengths, and angles to test the resin cast's removability from the silicone molds (Fig. 4). Currently, we constrain the maximum

ALGORITHM 1 Procedural Generation of Crystal Clusters

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1:  $\mathcal{O}^L \leftarrow \emptyset$   $\triangleright$  container for large crystals
2:  $\mathcal{P} \leftarrow \text{SAMPLINGONSURFACE}(\mathcal{S}, N_{\text{crystals}})$ 
3: if target_shape_specified? then
4:    $\mathcal{Q} \leftarrow \text{SAMPLINGONTARGETSHAPE}(\mathcal{M}, N_{\text{crystals}})$ 
5: for  $i = 1 \rightarrow N_{\text{crystals}}$  crystals do
6:    $p_i \leftarrow \mathcal{P}[i]$ 
7:   if target_shape_specified? then
8:      $q_i \leftarrow \mathcal{Q}[i]$ ;  $v_i \leftarrow p_i - q_i$ ;  $\theta_{\text{crystal}} \leftarrow v_i.\text{growth\_dir\_angle}$ 
9:      $h, t \leftarrow \text{CALCLENGTH}(h_{\text{crystal}}, t_{\text{crystal}}, \sigma_{\text{length}}, |v_i|)$ 
10:     $d \leftarrow \text{CALCDIRECTION}(\theta_{\text{crystal}}, \theta_{\max})$ 
11:   else
12:      $h, t \leftarrow \text{CALCLENGTH}(h_{\text{crystal}}, t_{\text{crystal}}, \sigma_{\text{length}})$ 
13:      $d \leftarrow \text{CALCDIRECTION}(\theta_{\text{crystal}}, \theta_{\max}, \sigma_{\text{direction}})$ 
14:      $r \leftarrow \text{CALCRADIUS}(r_{\text{crystal}}, \sigma_{\text{radius}}, i, \mathcal{P})$ 
15:      $o \leftarrow \text{CREATECRYSTAL}(h, r, t, \text{offset}).\text{transform}(p_i, d, s_{\text{crystal}})$ 
16:      $\mathcal{O}^L \leftarrow \mathcal{O}^L \cup o$ 
17:    $\mathcal{O}^M \leftarrow \text{GENERATEMIDCRYSTALS}(\mathcal{O}^L, \mathcal{S}, \phi_{m,1}, \sigma_{\text{overlap}}, \rho_{\text{crystal}})$ 
18:    $\mathcal{O}^S \leftarrow \text{GENERATESMALLCRYSTALS}(\mathcal{O}^L \cup \mathcal{O}^M, \mathcal{S}, \phi_{s,m}, \sigma_{\text{overlap}})$ 
19:    $\mathcal{O}^{\text{all}} \leftarrow \mathcal{O}^L \cup \mathcal{O}^M \cup \mathcal{O}^S$   $\triangleright$  merge all crystals

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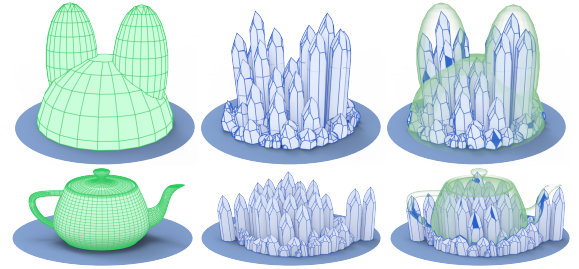


Figure 3: From left to right: target shapes, generated models, and overlaid. Top row: a successfully approximated crystal cluster. Bottom row: a failure case.

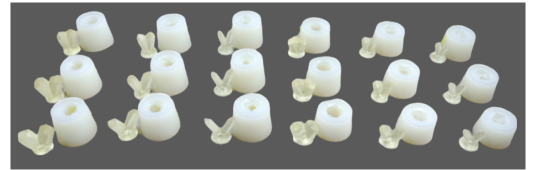


Figure 4: Silicone molds for resin cast.

overhang angle of procedurally generated cluster models by setting the identified value to $\theta_{\max}$ by our physical experiments. In the future, we would like to automatically predict removability using simulation or a data-driven method.

REFERENCES

- J. D. H. Donnay and Y. Le Page. 1978. The vicissitudes of the low-quartz crystal setting or the pitfalls of enantiomorphism. *Acta Crystallographica Section A* 34, 4 (Jul 1978), 584–594. <https://doi.org/10.1107/S0567739478001242>
- SideFX. 2022. *Houdini*. Retrieved Aug 1, 2022 from <https://www.sidefx.com/products/houdini/>
- Jerry O. Talton, Yu Lou, Steve Lesser, Jared Duke, Radomir Měch, and Vladlen Koltun. 2011. Metropolis Procedural Modeling. *ACM Trans. Graph.* 30, 2, Article 11 (April 2011), 14 pages. <https://doi.org/10.1145/1944846.1944851>